Lectures in Mathematics

Essential Math for Physics Undergraduates

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About the book

Essential Maths for Physics Undergrads aims to introduce basic, and crucial mathematical areas and tools required for any physicist. No technical knowledge is assumed past the point of high-school science and mathematics. This book may serve as a pre-university text for those who will later go on to study a scientific discipline (not necessarily just maths and physics, although some areas may be most useful to students of these subjects) or as a semi-technical source for curious readers.

The layout of the book will be similar to what you might find in a university-level mathematics module. Concepts will be introduced with clear definitions via **Definition**, **Theorem**, **Corollary**... etc... markers. This may also help students who are not yet acquainted with such conventions to get used to them before they begin their university studies.

Each chapter will include many different examples along with the above mentioned flags. Some practice questions will also be included at the end of each chapter. These will, however, be limited.

Chapters in this book may be used separately from each other as there will be minimal overlap in material. It should be said, however, that titles of all chapters should be mentally prefixed with 'Introductory' as no advanced concepts are covered.

Keywords, Symbols and Shorthands

s.t. (or $|$) = "such that"
iff = "if and only if"
$\in$ = "is a member of..."
$\forall$ = "for all"
$\exists$ = "there exists"
$\mathbb{N}$ = "the set of all natural numbers; 1, 2, 3, 4, 5, ..."
$\subseteq$ = "is a subset of..."

Contents

Chapter 1

Linear Algebra

Introduction

Linear Algebra is fundamental to most applied mathematics. It allows us to work in multiple dimensions and to reduce calculations using simultaneous computations. It's particularly useful when using coordinates and consequently appears in various scientific studies. We will look first at fundamental operations in vectors, then matrices, and then pivot to key knowledge of matrices.

1.1 Vectors

1.1.1 Basic information

The number of components in vectors and matrices (their *dimension*) reflects what dimension they are in. A vector with two components is operating on a plane; a matrix with three rows and three columns is operating in a 3D field.

Definition: *(The nth dimension)* $\mathbb{R}^n$, where $n \in \mathbb{N}$, represents the n^{th} dimension.

Example:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 \tag{1.1}$$

We will typically operate in $n = 2$ (representing a plane) and $n = 3$ (ideal for 'real-life' scenario), but we will refer to the n^{th} dimension for generality.

Definition: *(An n-by-m matrix)* For a "n-by-m" (or $n \times m$) matrix A, A has n rows and m columns, and the given elements are commonly referred to as A_{nm}. If the elements of A belong to a vector-space V then we say that $A \in V^{n\times m}$.

Example: In the following matrix, $A_{12} = 2$, $A_{34} = 12$, $A_{33} = 11$

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix} \tag{1.2}$$

and since all elements of A are in $\mathbb{R}$, we say that $A \in \mathbb{R}^{3\times 4}$.

Definition: *(Row and column vectors)* A n x m matrix with $n = 1$ is a *row vector* with m elements. A n x m matrix with $m = 1$ is a *column vector* with m elements.

Example:

$\mathbf{v} = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}$ is a row vector. $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$ is a column vector.

Corollary: *(The transpose)* The *transpose* (denoted by a superscript $\mathbb{T}$) of a row vector is a column vector with the same values, ordered identically from top to bottom as in the original from left to right. Similarly, the transpose of a column vector is a row vector also maintaining the original order of components when read left to right for the row vector, top to bottom in the column vector.

Example: For $\mathbf{v} = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}$,

$$\mathbf{v}^{\mathbb{T}} = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}^{\mathbb{T}} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \tag{1.3}$$

$$\mathbf{v}^{\mathbb{T}^{\mathbb{T}}} = \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}^{\mathbb{T}} = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \tag{1.4}$$

Remark: We will often use the transpose for in-line vectors to keep writing kempt.

Every vector and matrix exists with respect to a basis. We will discuss this more in section ?? but for now it is enough to assume that, unless stated otherwise, all following matrices and vertices are with respect to the *standard basis*:

Definition: *(The standard basis)* For $\mathbf{v} \in \mathbb{R}^n$, the standard basis is the set of n vectors, where for the n^{th} vector, its n^{th} component is equal to 1, and all other components are 0.

Example: The standard basis in $\mathbb{R}^3$ consists of the following vectors;

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, and \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{1.5}$$

1.1.2 Vector operations

We will now look at vector operations; addition, subtraction, scalar multiplication and so on.

Vector addition and subtraction

Two or more vectors can only be added (or subtracted) if they are in the same vector space (to be explained later; for now assume all are with respect to the standard basis and therefore in the n^{th}

dimension (which is an example of a vector space - see section ?? for more information.)

Definition: *(Vector addition)* For vertices **v** and **w**, with elements $v_1, v_2, v_3, ..., v_n$ and $w_1, w_2, w_3, ..., w_n$ respectively, and with both in respect to the vector space $\mathbf{V} \subseteq \mathbb{R}^{\ltimes}$, (in notation; for **v**,**w** $\in$ **V**, where $\mathbf{v} = v_1, v_2, v_3, ..., v_n$ and $\mathbf{w} = w_1, w_2, w_3, ..., w_n$),

$$\mathbf{v} + \mathbf{w} = [v_1, v_2, v_3, ..., v_n] + [w_1, w_2, w_3, ..., w_n] = [v_1 + w_1, v_2 + w_2, v_3 + w_3, ..., v_n + w_n] \tag{1.6}$$

Example:

$$[1, 2, 3, 4, 5] + [6, 7, 8, 9, 10] = [7, 9, 11, 13, 15] \tag{1.7}$$

Scalar multiplication

We can multiply a vector by a scaler simply by multiplying each component by that scalar.

Definition: *(Scalar multiplication of a vector)* For $\mathbf{v} \in \mathbf{V}$, where $\mathbf{V} \subseteq \mathbb{R}^{\ltimes}$ is some vector space and $\mathbf{v} = [v_1, v_2, v_3, ..., v_n]$, and for $\lambda \in \mathbb{R}$,

$$\lambda\mathbf{v} = \lambda[v_1, v_2, v_3, ..., v_n] = [\lambda v_1, \lambda v_2, \lambda v_3, ..., \lambda v_n] \tag{1.8}$$

Example:

$$10 \cdot [1, 2, 3, 4, 5] = [10, 20, 30, 40, 50] \tag{1.9}$$

Corollary: *(Collinear vectors)* Two vectors **v** and **w** are *collinear* when they are parallel or lie along the same line; in other words, when $\mathbf{v} = \lambda\mathbf{w}$ for some $\lambda \in \mathbb{R}$.

Example:
$[1, 2, 3, 4, 5]$ and $[10, 20, 30, 40, 50]$ are collinear as $10 \cdot [1, 2, 3, 4, 5] = [10, 20, 30, 40, 50]$

The dot product

When considering multiplication and vectors, there are three relevant operations: scalar multiplication, the *dot product*, and the *cross product*. We have already discussed scalar multiplication, and we now pivot to the dot product, which can be considered indicative of the 'angle' between two vectors, but more on this later.

Definition: *(The dot product)* Given two vectors $\mathbf{v} = [v_1, v_2, v_3, ..., v_n]$ and $\mathbf{w} = [w_1, w_2, w_3, ..., w_n]$, with **v**,**w** $\in \mathbb{V}$, for some vector space $\mathbb{V} \subseteq \mathbb{R}^n$, the *dot product* is defined as

$$\mathbf{v} \cdot \mathbf{w} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \dots \\ v_n \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \dots \\ w_n \end{bmatrix} = \sum_{i=1}^{n} v_i w_i \tag{1.10}$$

Very important note: the result of the dot product is a scalar; if one attains a vector as a result of the dot product computation, one has erred.

Example:

$$\begin{bmatrix}1\\2\\3\\4\\5\\6\\7\end{bmatrix} \cdot \begin{bmatrix}1\\2\\3\\4\\5\\6\\7\end{bmatrix} = 1+4+9+16+25+36+49 = 140 \tag{1.11}$$

Corollary: *(Dot product = 0)* If $\mathbf{v} \cdot \mathbf{w} = 0$ (for $\mathbf{v}$, $\mathbf{w} \in \mathbb{V}$ and some vector space $\mathbb{V} \subseteq \mathbb{R}^n$), then the vectors $\mathbf{v}$ and $\mathbf{w}$ are perpendicular.

Example:

$$\begin{bmatrix}1\\-2\\3\end{bmatrix} \cdot \begin{bmatrix}1\\\frac{1}{2}\\0\end{bmatrix} = 1-1=0, \tag{1.12}$$

Hence $\begin{bmatrix}1 & -2 & 3\end{bmatrix}^{\mathbb{T}}$ and $\begin{bmatrix}1 & \frac{1}{2} & 0\end{bmatrix}^{\mathbb{T}}$ are perpendicular.

The cross product

Our final multiplication operation is the *cross product*; it provides a third vector that is perpendicular to both initial vectors. It can only be used in $\mathbb{R}^3$ and $\mathbb{R}^7$, and we will only look at $\mathbb{R}^3$.

Definition: *(The cross product (in 3D))* Given two vectors $\mathbf{v} = [v_1, v_2, v_3]$ and $\mathbf{w} = [w_1, w_2, w_3]$, with $\mathbf{v}$,$\mathbf{w} \in \mathbb{V}$, for some vector space $\mathbb{V} \subseteq \mathbb{R}^3$, the *cross product* is defined as

$$\mathbf{v} \times \mathbf{w} = \begin{bmatrix}v_1\\v_2\\v_3\end{bmatrix} \times \begin{bmatrix}w_1\\w_2\\w_3\end{bmatrix} = \begin{bmatrix}v_2w_3 - v_3w_2\\v_3w_1 - v_1w_3\\v_1w_2 - v_2w_1\end{bmatrix} \tag{1.13}$$

Example:

$$\begin{bmatrix}1\\-2\\3\end{bmatrix} \times \begin{bmatrix}4\\1\\2\end{bmatrix} = \begin{bmatrix}-4-3\\12-2\\1-(-8)\end{bmatrix} = \begin{bmatrix}-7\\10\\9\end{bmatrix} \tag{1.14}$$

Hence, $\begin{bmatrix}-7 & 10 & 9\end{bmatrix}^{\mathbb{T}}$ is perpendicular to $\begin{bmatrix}1 & -2 & 3\end{bmatrix}^{\mathbb{T}}$ and $\begin{bmatrix}4 & 1 & 2\end{bmatrix}^{\mathbb{T}}$ (you are welcome to check this using the dot product!)

Corollary: *(The right hand rule)* You may have realised that any two vectors actually have two mutually perpendicular vectors; if one has $\mathbf{v} \times \mathbf{w} = \mathbf{z}$, then the other perpendicular vector is $-\mathbf{z}$.

Cheat method to the cross product

Using the definition is useful but often difficult to remember; there is an easier method.

Given two column vectors, start by covering up the 1st value in both vectors (the 'top' ones). This should leave the two bottom values from both vectors visible. Next use a '$\times$'; from these four values, multiply the top left and bottom right, then subtract the top right multiplied by the bottom left. You have now found your 1st value for your resulting vertex.

Example: Consider the following vectors from before: $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$. To find the cross product, start by covering the top row; this leaves $\begin{bmatrix} -2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Hence perform the operation $(-2) \cdot 2 - 1 \cdot 3 = -7$. This is the top value of the resulting vector.

Next, start by cover up the 2nd row in both vectors (the 'middle' ones). Next use a '×', like with the first row, but take the negative of the result; from these four values, multiply the top right by the bottom left , then subtract the top left multiplied by the bottom right. You have now found your 2nd value for your resulting vertex.

Example (continued): Consider again the following vectors from before: $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$

Cover the middle row; this leaves $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 2 \end{bmatrix}$. Hence perform the operation $(4) \cdot 3 - 1 \cdot 2 = 10$. This is the middle value of the resulting vector.

Finally, cover up the 3rd row in both vectors (the 'bottom' ones). Next use a '×', like with the first step; from these four values, multiply the top left and bottom right, then subtract the top right multiplied by the bottom left. You have now found your final value for your resulting vertex.

Example (final): Consider once more the following vectors: $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$. Cover the bottom row; this leaves $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$. Hence perform the operation $1 \cdot 1 - 4 \cdot (-2) = 9$. This is the bottom value of the resulting vector.

Like in our earlier example, $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} \times \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 - 3 \\ 12 - 2 \\ 1 - (-8) \end{bmatrix} = \begin{bmatrix} -7 \\ 10 \\ 9 \end{bmatrix}$

1.1.3 Vector spaces

Span of vectors

The *span* of a vector is the set of all possible vectors that could be created by multiplying that vector by some scalar. The span of a *set* of vectors is the set of all possible vectors that could be made by combining them linearly.

Definition:*(Span)* Given a subset of $\mathbb{R}^n$ with j vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, ..., \mathbf{v}_j \in V$, $V \in \mathbb{R}^n$, its *span* is the set

$$\text{Span}(V) = \left\{ \sum_{i=1}^{j} \alpha_i \mathbf{v}_i \text{ s.t. } \alpha_j \in \mathbb{R}, \forall j \right\} \tag{1.15}$$

i.e. it is the set of all linear combinations of vectors in the subset.

Example: Consider the standard basis in 3D: $\begin{bmatrix}1 & 0 & 0\end{bmatrix}^{\mathbb{T}}$, $\begin{bmatrix}0 & 1 & 0\end{bmatrix}^{\mathbb{T}}$, and $\begin{bmatrix}0 & 0 & 1\end{bmatrix}^{\mathbb{T}}$. These span $\mathbb{R}^3$; you can create any 3D vector using some linear combination of them. For example,

$$\begin{bmatrix}5\\21.2\\-41\end{bmatrix} = 5 \cdot \begin{bmatrix}1\\0\\0\end{bmatrix} + 21.2 \cdot \begin{bmatrix}0\\1\\0\end{bmatrix} + (-41) \cdot \begin{bmatrix}0\\0\\1\end{bmatrix} \tag{1.16}$$

Linear dependency of vectors

When looking at a set of vectors that span a given vector space, it is possible to have redundant vectors within this set. For example, parallel vectors both span the same set of possible vectors; one term we can use to avoid this is *(linear independence)*

Definition:*(Linear independence)* A set of j vectors, $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, ..., \mathbf{v}_j \in V$, $V \in \mathbb{R}^n$, is considered linearly independent if and only if the only solution to the following equation is $\alpha_1 = \alpha_2 = \alpha_3 = ... = \alpha_j = 0$:

$$\sum_{i=1}^{j} \alpha_i \mathbf{v}_i = \mathbf{0}, \text{ s.t. } \alpha_j \in \mathbb{R}, \forall j \tag{1.17}$$

i.e. There is no linear combination for non-zero α_j that is equal to 0.

Example: $\begin{bmatrix}1 & 0 & 4 & 5\end{bmatrix}$ and $\begin{bmatrix}5 & 0 & 20 & 25\end{bmatrix}$.

$$\begin{bmatrix}1 & 0 & 4 & 5\end{bmatrix} - 5 \cdot \begin{bmatrix}5 & 0 & 20 & 25\end{bmatrix} = \mathbf{0} \tag{1.18}$$

.

Hence, $\begin{bmatrix}1 & 0 & 4 & 5\end{bmatrix}$ and $\begin{bmatrix}5 & 0 & 20 & 25\end{bmatrix}$ are linearly dependent.

Example: Consider the standard basis in 3D, in row vector form: $\begin{bmatrix}1 & 0 & 0\end{bmatrix}$, $\begin{bmatrix}0 & 1 & 0\end{bmatrix}$, and $\begin{bmatrix}0 & 0 & 1\end{bmatrix}$.

If $\alpha_1 \begin{bmatrix}1 & 0 & 0\end{bmatrix} + \alpha_2 \begin{bmatrix}0 & 1 & 0\end{bmatrix} + \alpha_3 \begin{bmatrix}0 & 0 & 1\end{bmatrix} = \mathbf{0}$, then $\alpha_1 = \alpha_2 = \alpha_3 = 0$, as $\alpha_1 + 0 + 0 = 0$ only if $\alpha_1 = 0$, $0 + \alpha_2 + 0 = 0$ only if $\alpha_2 = 0$, and $0 + 0 + \alpha_3 = 0$ only if $\alpha_3 = 0$.

Hence, the standard basis is linearly independent.

Bases and vector spaces

Definition:*(Vector spaces)* A *vector space* V is a set of vectors that adheres to all of the following axioms: for $\mathbf{v}, \mathbf{w}, \mathbf{x} \in V$, $\alpha, \beta \in \mathbb{R}$,

1. Closure under addition: $\mathbf{v} + \mathbf{w} \in V$.
2. Commutativity: $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$.
3. Associativity under addition: $(\mathbf{v} + \mathbf{w}) + \mathbf{x} = \mathbf{v} + (\mathbf{w} + \mathbf{x})$.
4. Additive identity: $\exists\ \mathbf{0} \in V$ s.t. $\mathbf{v} + \mathbf{0} = \mathbf{v}$.
5. Additive inverse: $\exists\ (-\mathbf{v}) \in V$ s.t. $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.
6. Closure under multiplication: $(\alpha \cdot \mathbf{v}) \in V$.
7. Distributivity 1: $\alpha \cdot (\mathbf{v} + \mathbf{w}) = \alpha \cdot \mathbf{v} + \alpha \cdot \mathbf{w}$.
8. Distributivity 2: $(\alpha + \beta) \cdot \mathbf{v} = \alpha\mathbf{v} + \beta\mathbf{v}$.
9. Associativity under multiplication: $\alpha(\beta \cdot \mathbf{v}) = (\alpha\beta) \cdot \mathbf{v}$.
10. Multiplication: $1 \cdot \mathbf{v} = \mathbf{v}$.

Example: $\mathbb{R}^3$ is a vector space; you are welcome to check that it fulfills all the above axioms!

A *basis* is a set of linearly independent vectors that span a vector space; in other words, it's a set of vectors that can be combined to attain any vector in the vector space.

Definition: *(The basis)* A set of vectors $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, ..., \mathbf{v}_j\}, j \in \mathbb{N}$, forms a *basis* of a vector space $V \subseteq \mathbb{R}^n$ if and only if B spans V;

$$\text{Span}(B) = \left\{\sum_{i=1}^{j} \alpha_i \mathbf{v}_i \text{ s.t. } \alpha_j \in \mathbb{R}, \forall j\right\} = \text{Span}(V) \tag{1.19}$$

and $\mathbf{v}_j$ are linearly independent:

$$\sum_{i=1}^{j} \alpha_i \mathbf{v}_i = \mathbf{0}, \text{ s.t. } \alpha_j \in \mathbb{R}, \forall j, \text{ if and only if all } \alpha_j = 0 \tag{1.20}$$

Example: We have already established that the standard *basis* spans $\mathbb{R}^3$ and is linearly independent; as suggested by its name, it forms a basis of $\mathbb{R}^3$. However, it isn't the only basis for $\mathbb{R}^3$. You could use the basis $\left\{\begin{bmatrix}1 & 1 & 0\end{bmatrix}, \begin{bmatrix}1 & 0 & 1\end{bmatrix}, \begin{bmatrix}0 & 1 & 1\end{bmatrix}\right\}$: For α_i, $i = 1, 2, 3$, set $\alpha_1 \begin{bmatrix}1 & 1 & 0\end{bmatrix} + \alpha_2 \begin{bmatrix}1 & 0 & 1\end{bmatrix} + \alpha_3 \begin{bmatrix}0 & 1 & 1\end{bmatrix} = \mathbf{0}$. Then, looking only at the first entries in the vectors, $\alpha_1 + \alpha_2 + 0 = 0$, hence $\alpha_1 = -\alpha_2$. Looking only at the second entries, $\alpha_1 + 0 + \alpha_3 = 0$, hence $\alpha_1 = -\alpha_3$. Looking now at only the final entries in the vectors, $0 + \alpha_2 + \alpha_3 = 0$. Hence, $\alpha_2 = -\alpha_3$. Combining these, we have that $\alpha_1 = -\alpha_1$, hence $\alpha_1 = 0$. From this, $\alpha_2 = \alpha_3 = 0$ as well. Hence the only solution to the above is $\alpha_i = 0, \forall i$. Hence our new basis is linearly independent.

There is a more complicated method of checking whether a set spans a given vector space, which we will discuss later, but for now it is sufficient to note that:

$$\begin{bmatrix}1 & 0 & 0\end{bmatrix} = \frac{1}{2}(\begin{bmatrix}1 & 1 & 0\end{bmatrix} + \begin{bmatrix}1 & 0 & 1\end{bmatrix} - \begin{bmatrix}0 & 1 & 1\end{bmatrix}) \tag{1.21}$$

$$\begin{bmatrix}0 & 1 & 0\end{bmatrix} = \frac{1}{2}(\begin{bmatrix}1 & 1 & 0\end{bmatrix} + \begin{bmatrix}0 & 1 & 1\end{bmatrix} - \begin{bmatrix}1 & 0 & 1\end{bmatrix}) \tag{1.22}$$

$$\begin{bmatrix}0 & 0 & 1\end{bmatrix} = \frac{1}{2}(\begin{bmatrix}1 & 0 & 1\end{bmatrix} + \begin{bmatrix}0 & 1 & 1\end{bmatrix} - \begin{bmatrix}1 & 1 & 0\end{bmatrix}) \tag{1.23}$$

In other words, each element of the standard basis can be written as a linear combination of our basis. As we know that the standard basis spans $\mathbb{R}^3$, our basis also spans $\mathbb{R}^3$.

Remark: A vector with respect to the standard basis can be expressed using that basis; for example, $\begin{bmatrix}5 & 3 & 1\end{bmatrix} = 5 \cdot \begin{bmatrix}1 & 0 & 0\end{bmatrix} + 3 \cdot \begin{bmatrix}0 & 1 & 0\end{bmatrix} + 1 \cdot \begin{bmatrix}0 & 0 & 1\end{bmatrix}$. Similarly, you can express a vector with respect to a different basis, provided that vector is in the vector space the basis is for. For example: $\begin{bmatrix}5 & 3 & 1\end{bmatrix} = 3 \cdot \begin{bmatrix}1 & 1 & 0\end{bmatrix} + 2 \cdot \begin{bmatrix}1 & 0 & 1\end{bmatrix} + (-1) \begin{bmatrix}0 & 1 & 1\end{bmatrix}$. You do not currently need to know how to find the constants required to do so (but it will be discussed later).

Vector algebra

We've seen in our determining of linear-dependency that we can find simultaneous equations for our constants. Fortunately, with simple values, we can often find this easily enough. However, when we do not know all the entry values of a vector, or have more variables, this can get complicated quickly;

consequently, we often use matrices to keep our working clear and aid us with solving problems.

Example: Let's say we wanted to find a basis $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ for $\mathbb{R}^3$, but that it **had** to fulfill the following properties:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 5 \\ 10 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ v_{23} \\ v_{33} \end{bmatrix} \text{ for some } v_{23}, v_{33} \in \mathbb{R} \tag{1.24}$$

, and

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3 = \begin{bmatrix} 65 \\ 10 \\ 3 \end{bmatrix}, \text{ for some } \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R} \tag{1.25}$$

. How would we condense this information?

We would, for the purposes of this demonstration, use a matrix (as well as some vectors):

The following expression contains all this information; the matrix is made up of our column vectors, and is being multiplied by a vector containing constants (that in practice is the linear combination of our basis vectors). $\mathbf{v}_2$'s elements here are written as v_{12}, v_{22}, v_{32}, where $v_{12}, v_{22}, v_{32} \in \mathbb{R}$

$$\begin{bmatrix} 1 & v_{12} & 0 \\ 5 & v_{22} & v_{23} \\ 10 & v_{32} & v_{33} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} 65 \\ 10 \\ 3 \end{bmatrix} \tag{1.26}$$

Remark: We often used *augmented matrix* notation: When writing the above repeatedly for matrix operations, it can get tedious. Consequently, when we have the above structure of a matrix multiplied by a vector with all unkown entries, equal to a vector, we can use the following shorthand:

$$\left[\begin{array}{ccc|c} 1 & v_{12} & 0 & 65 \\ 5 & v_{22} & v_{23} & 10 \\ 10 & v_{32} & v_{33} & 3 \end{array}\right] \tag{1.27}$$

To be clear, these are still in practice a 3 by 3 matrix, multiplied by a vector with unknown entries and equal to the right hand column vector. This is simply a common shorthand.

Now that we have seen an example of where matrices can be useful, we can start to learn their qualities.

1.2 Matrices

1.2.1 The Identity Matrix

Let's start with the simplest matrix possible; the identity matrix. This matrix is formed using the standard basis as its columns.

Definition:*(The identity matrix)* The identity matrix in the n^{th} dimension, $\mathbb{I}^n$, is a $n \times n$ matrix, where the values on the main diagonal are each 1, and all other values are 0.

Example: The following matrix is the identity matrix ($\mathbb{I}$) for $\mathbb{R}^{3\times3}$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{1.28}$$

1.2.2 Matrix operations

Matrix addition and subtraction

Similarly to vectors, matrices can only be added or subtracted if they have identical dimensions.

Definition: *(Matrix addition)* two $m \times n$ matrices, where $m, n \in \mathbb{R}$, and with values a_{ij} and b_{kl}, where i and k correspond to which row the values are found in, and j and l correspond to which column the values are found in, are added together by adding the corresponding entries together; $a_{ij} + b_{ij}$.

In other words, the values in the same 'position' get added together.

Example:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} + \begin{bmatrix} 2 & 3 & 4 \\ -5 & -3 & -1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+2 & 2+3 & 3+4 \\ 4-5 & 5-3 & 6-1 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 3 & 5 & 7 \\ -1 & 2 & 5 \\ 7 & 8 & 10 \end{bmatrix} \tag{1.29}$$

Scalar multiplication

To multiply a matrix by a scaler, you multiply each matrix entry by that scalar.

Definition: *(Scalar multiplication of a matrix)* For a matrix $\mathbf{M} \in \mathbb{R}^{m\times n}$, with values a_{ij} for $1 \leq i \leq m, 1 \leq j \leq n$, $a_{ij} \in \mathbb{R}\forall i, j$ and a value $\lambda \in \mathbb{R}$,

$$\lambda \cdot \mathbf{M} = \lambda \begin{bmatrix} a_{11} & a_{12} & ... & a_{1m} \\ a_{21} & a_{22} & ... & a_{2m} \\ ... & ... & ... & ... \\ a_{n1} & ... & ... & ... \end{bmatrix} = \begin{bmatrix} \lambda a_{11} & \lambda a_{12} & ... & \lambda a_{1m} \\ \lambda a_{21} & \lambda a_{22} & ... & \lambda a_{2m} \\ ... & ... & ... & ... \\ \lambda a_{n1} & ... & ... & ... \end{bmatrix} \tag{1.30}$$

Example:

$$4 \cdot \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 12 \\ 16 & 20 & 24 \\ 28 & 32 & 36 \end{bmatrix} \tag{1.31}$$

Elementary row operations and Reduced row echelon form

We can perform *elementary row operations* to get a matrix or system to *reduced row echelon form*; this is often helpful for finding solutions to linear systems. There are three operations we can use:

1. Row addition (adding one row to another)
2. Row swapping (swapping rows)
3. Scalar multiplication (excluding multiplication by 0)

When performing elementary row reduction, we start by getting the first diagonal entry to 1. We then use this top row to get all the values vertically below our leading diagonal (all the entries of the first column except the top one) to 0. We then repeat this process for each other diagonal entry.

Example: We begin with the following matrix:

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \tag{1.32}$$

We want the top left value, the leading diagonal, to be 1, so we swap the first and second row:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & \frac{1}{2} \end{bmatrix} \tag{1.33}$$

The values beneath this are now already 0, so we consider the second diagonal; this is already 1, so we consider the value below it. As this is not 0, we subtract the second row from the bottom row; this renders the value below the second diagonal as 0.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -\frac{1}{2} \end{bmatrix} \tag{1.34}$$

Finally, we multiply the bottom row by -2, and we have arrived at reduced row echelon form:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \tag{1.35}$$

A more difficult example. **Example:** Let's reduce the following matrix:

$$\begin{bmatrix} -33 & 12 & 28 & 31 \\ 45 & 42 & -7 & 6 \\ 1 & -22 & \frac{1}{2} & \frac{177}{10} \\ -\frac{391}{10} & -\frac{78}{10} & \frac{198}{10} & -\frac{424}{10} \end{bmatrix} \tag{1.36}$$

We begin by multiplying the first row by $-\frac{1}{33}$:

$$\begin{bmatrix} 1 & -\frac{4}{11} & -\frac{28}{33} & -\frac{31}{33} \\ 45 & 42 & -7 & 6 \\ 1 & -22 & \frac{1}{2} & \frac{177}{10} \\ -\frac{391}{10} & -\frac{78}{10} & \frac{198}{10} & -\frac{424}{10} \end{bmatrix} \tag{1.37}$$

Next, we subtract the top row, multiplied by 45, from the second row. This reduces the entry below our diagonal to 0.

$$\begin{bmatrix} 1 & -\frac{4}{11} & -\frac{28}{33} & -\frac{31}{33} \\ 0 & \frac{642}{11} & \frac{343}{11} & \frac{531}{11} \\ 1 & -22 & \frac{1}{2} & \frac{177}{10} \\ -\frac{391}{10} & -\frac{78}{10} & \frac{198}{10} & -\frac{424}{10} \end{bmatrix} \tag{1.38}$$

We now want the third row's first value to also be 0, so we subtract the first row from the third row:

$$\begin{bmatrix} 1 & -\frac{4}{11} & -\frac{28}{33} & -\frac{31}{33} \\ 0 & \frac{642}{11} & \frac{343}{11} & \frac{531}{11} \\ 0 & -\frac{238}{11} & \frac{89}{66} & \frac{6151}{330} \\ -\frac{391}{10} & -\frac{78}{10} & \frac{198}{10} & -\frac{424}{10} \end{bmatrix} \tag{1.39}$$

Next we add to our bottom row $\frac{391}{10}$ multiplied by our top row, to get all but the top value of our first column to 0:

$$\begin{bmatrix} 1 & -\frac{4}{11} & -\frac{28}{33} & -\frac{31}{33} \\ 0 & \frac{642}{11} & \frac{343}{11} & \frac{531}{11} \\ 0 & -\frac{238}{11} & \frac{89}{66} & \frac{6151}{330} \\ 0 & -\frac{167939}{21505} & \frac{1275997}{64515} & -\frac{2736986}{64515} \end{bmatrix} \tag{1.40}$$

Now we want our second diagonal value to be equal to 1, so we multiply our second row by $\frac{11}{642}$:

$$\begin{bmatrix} 1 & -\frac{4}{11} & -\frac{28}{33} & -\frac{31}{33} \\ 0 & 1 & \frac{343}{642} & \frac{177}{214} \\ 0 & -\frac{238}{11} & \frac{89}{66} & \frac{6151}{330} \\ 0 & -\frac{167939}{21505} & \frac{1275997}{64515} & -\frac{2736986}{64515} \end{bmatrix} \tag{1.41}$$

We now want the values below this to be 0; we add $\frac{11}{238}$ multiplied by the second row to our third row, and $\frac{21505}{167939}$ multiplied by our second row to our fourth row.

$$\begin{bmatrix} 1 & -\frac{4}{11} & -\frac{28}{33} & -\frac{31}{33} \\ 0 & 1 & \frac{343}{642} & \frac{177}{214} \\ 0 & 0 & \frac{22367}{21828} & \frac{6151}{330} \\ 0 & 0 & \frac{15338870927579}{772867033730} & -\frac{98118823128481}{2318601101190} \end{bmatrix} \tag{1.42}$$

Next, in order to have a 1 on the diagonal, we multiply the third row by $\frac{21828}{22367}$:

$$\begin{bmatrix} 1 & -\frac{4}{11} & -\frac{28}{33} & -\frac{31}{33} \\ 0 & 1 & \frac{343}{642} & \frac{177}{214} \\ 0 & 0 & 1 & \frac{22377338}{1230185} \\ 0 & 0 & \frac{15338870927579}{772867033730} & -\frac{98118823128481}{2318601101190} \end{bmatrix} \tag{1.43}$$

Once again we reduce the value below this to 0:

$$\begin{bmatrix} 1 & -\frac{4}{11} & -\frac{28}{33} & -\frac{31}{33} \\ 0 & 1 & \frac{343}{642} & \frac{177}{214} \\ 0 & 0 & 1 & \frac{22377338}{1230185} \\ 0 & 0 & 0 & \frac{150354981982848988041306464 9}{35564723023695946140719010} \end{bmatrix} \tag{1.44}$$

Finally, we multiply the final row by $\frac{35564723023695946140719010}{1503549819828489880413064649}$, and we have found the reduced echelon form!

$$\begin{bmatrix} 1 & -\frac{4}{11} & -\frac{28}{33} & -\frac{31}{33} \\ 0 & 1 & \frac{343}{642} & \frac{177}{214} \\ 0 & 0 & 1 & \frac{22377338}{1230185} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1.45}$$

Carla's to do: - add wordy bits

12. matrix multiplication (AB =/ BAcommutative) associative and distributive. premultiply post-multiply? 13. matrix to a power and 0 =identity matrix 14. inverse of matrices (invertible/singular) and deterimnates. inverse of inverse is matrix, inverse of product is inverse x inverse, unique

15. orthogonal, transpose 16. basis in terms of matrices and changing basis etc 17. eigenvalues and eigenvectors?

18. some other things: kroneker delta symmetric upper traingular etc

Chapter 2

Calculus

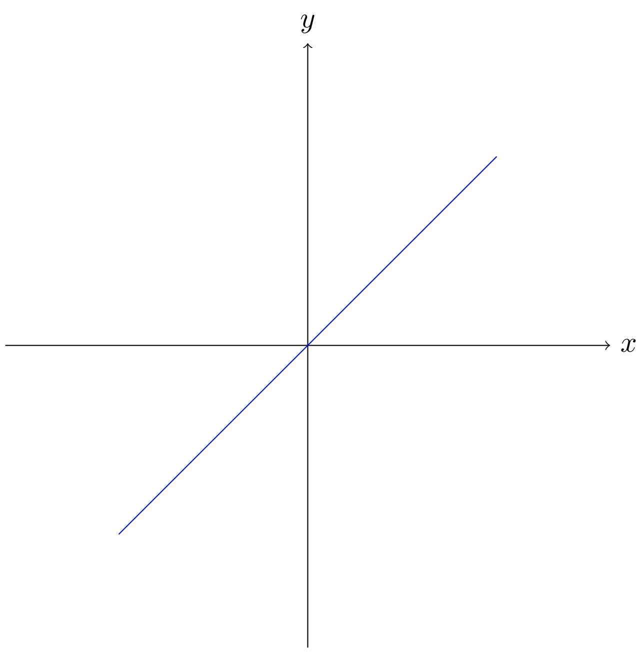
Introduction

This present chapter covers the infamous area of mathematics which arguably forms the very bedrock of modern physics. In fact, it indeed forms the basics of all modern hard sciences. Originally developed (or discovered) by Newton and Leibniz (full credit may be awarded to either party depending on your preference), Calculus is concerned, generally, with the study of continuous functions. In this chapter, we will first look at the idea of a limit (a way of evaluating functions at 'undefined' points). Then, we will use this new concept to define and use derivatives and integrals. On the way, we will also touch on the notion of operators which will be important in later chapters as well.

2.1 Properties of Functions

2.1.1 Continuity

First, we must establish the concept of continuity of functions. This concept is as crucial to later content as it is simple. In fact, the reader may already be well-acquainted with this concept without realizing. Consider the function $f(x) = x$. Is this a continuous function?



You probably would say yes without thinking about it. And, in fact, you'd be right. A function is considered continuous on an interval if it's well defined at every point in that interval.

Definition: *(Continuous functions)* Let f be a function on the interval $I = [a, b] \in \mathbb{R}$ s.t. $a < b$. Then, f is called continuous iff $f(x)$ is defined for all $x \in I$.

Example: Let $H : \mathbb{R} \to [0, 1]$ be the so-called Heaviside function

$$H(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases} \tag{2.1}$$

Since H is defined on the entirety of $\mathbb{R}$, we can say that H is continuous. (You may wish to plot H to double-check this claim.)

Example: Consider a variation of the Heaviside function $H^* : S \to [0, 1]$ where we define $S := [-\infty, 0] \cup [1, \infty]$. In this case, we write

$$H^*(x) = \begin{cases} 1 & \text{if } x \geq 1 \\ 0 & \text{if } x < 0 \end{cases} \tag{2.2}$$

Notice that the interval $[0, 1] \in \mathbb{R}$ is not covered by H^*. This means that $H^*(x)$ is undefined for $x \in [0, 1]$, and so, H^* is not continuous.

Remark: We will later see another important property, somewhat 'twin' to continuity, called smoothenss. However, it will be much easier to rigorously define this property once we have learned about derivatives. For now, a very hand-wavy definition of smoothness of a function is the lack of jagged points on the function. For example, if we take the function $f(x) = |x|$, we will see a sharp point at the origin, meaning it's not smooth.

2.1.2 Function Compositions

Composing functions is something which is not unfamiliar to high-school students. However, in this section, we will aim to define it in a more mathematically rigorous way. Let us start with a more complete definition of composition.

Definition: *(Composition)* Let f and g be two functions with domains X and Y respectively. A composition of f and g, $g \circ f$, is defined as

$$(g \circ f)(x) = g(f(x)) \tag{2.3}$$

for those $x \in X$ s.t. $f(x) \in Y$. There may be some $x \in X$ s.t. $f(x) \notin Y$. In such a case, the composition $g(f(x))$ is not defined. In addition, compositions are generally

- Non-commutative: $g \circ f \neq f \circ g$.
- Associative: $(f \circ g) \circ h = f \circ (g \circ h)$

Example: Consider the functions $f(x) = \sqrt{x + 1}$ and $g(x) = \cos(x)$ with domains $X = [-1, \infty)$ and $Y = \mathbb{R}$ respectively. Then, the co-domains (ranges) of f and g are $X^* = [0, \infty)$ and $Y^* = [-1, 1]$. Consequently,

$$(g \circ f)(x) = \cos(\sqrt{x+1}),\ \forall x \geq -1 \tag{2.4}$$
$$(f \circ g)(x) = \sqrt{\cos(x)+1},\ \forall x \in \mathbb{R} \tag{2.5}$$

Hence, $g \circ f \neq f \circ g$ in general.

Example: Consider the functions $f(x) = \sin(x)$, $g(x) = \sqrt{x}$ and $h(x) = x+3$. Hence, $((f \circ g) \circ h)(x) = \sin(\sqrt{h(x)})$. Similarly, $(f \circ (g \circ h))(x) = \sin((g \circ h)(x)) = \sin(\sqrt{h(x)})$. Hence, $(f \circ g) \circ h = f \circ (g \circ h)$.

2.2 Limits

2.2.1 Defining limits

We begin by defining a limit. You may have encountered functions in high-school maths which seemed to be undefined for certain values. If you have, there is a good chance that your teacher(s) simply told you to not worry about these values, or that these values are anomolous, or something along those lines. However, for some functions, you can define them at these 'anomolous' points. How? We employ limits...

As an example, consider the function below

$$f(x) = \frac{x^2 - 4}{x - 2} \tag{2.6}$$

Now, substitute in $x = 2$. You will notice that in this case, we get "0/0" which you would rightly say is nonsense (or undefined). However, you may be surprised to learn that this function is actually perfectly defined at $x = 2$. To see this, start substituting in values of x into $f(x)$ which are not 2, but rather are very close to 2. For example, substitute in the numbers 1.8, 1.9, 1.95, 1.99... and so on. You will notice that the answer you get starts to get closer and closer to (but never quite reaching) 4. This is in fact the basic concept of a limit. A bit more rigorously, we may define a limit in the following way.

Definition: *(Limit)* Let f be a function of a real variable $x \in \mathbb{R}$ and let $u \in \mathbb{R}$ be a point at which f is undefined. Then, the limit of f as x approaches u is

$$\lim_{x \to u} f(x) = L \tag{2.7}$$

where $L \in \mathbb{R}$. If f is continuos, then we can say that $L = f(u)$.

Remark: The above definition is a special case and in fact, limits can be defined for different vector spaces, not only $\mathbb{R}$.

Example: Consider the function

$$f(x) = \frac{1}{x} \tag{2.8}$$

Let's think about the value of $f(x)$ at infinitely large x. You will be quick to notice that it's once again undefined as $1/\infty$ is mathematical giberish. However, now let's consider

$$\lim_{x \to \infty} f(x) = \lim_{x \to \infty} \frac{1}{x} \tag{2.9}$$

the limit as x approaches ∞. For larger an larger x, $f(x)$ becomes smaller. In fact, for an arbitrarily large x value, the value of our function apporaches 0. Therefore, we can say with confidence that

$$\lim_{x\to\infty} \frac{1}{x} = 0 \tag{2.10}$$

Notice the subtle but crucial difference between saying that $1/\infty = 0$ and the eq.(2.5). We are not saying that division by ∞ is 0, because again, this doesn't make much sense for multiple reasons. One of them being that ∞ is not a number, so you cannot treat it as such. What we *are* saying, however, is that for larger and larger values of x this funcion $f(x) = 1/x$ *approaches* 0.

2.2.2 Undefined limits

Limits have proven quite a useful tool for evaluating functions at 'undefined' points. However, there are some functions which are not defined at particular points. That is, it's not always guaranteed that any limit of any function will be defined. These are called undefined limits or undefinable limits.

Example: Consider the function

$$f(x) = \sin x \tag{2.11}$$

How would we go about finding the value of f for an infinitely large value of x? Well, we would employ the usual strategy of trying to find the limit of f as x approaches infinity. However, sin is a periodic function. This means it never settles on a value as it extends to positive and negative infinity. It will always oscillate between $+1$ and -1. Therefore, we say that the limit

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \sin x \tag{2.12}$$

is undefined. Or that the limit doesn't exist.

2.2.3 One-sided limits

So far, we have been concerned with 'two-sided' limits. These are your standard limits for which it doesn't matter where you approach the limit point from. In every case, the limit will take on the same value. This makes intuitive sense. However, in some cases, we may encounter what's called a 'one-sided' limit. These are the exact opposite of a two-sided limit; that is, when you approach a limit point on a function, it may evaluate to different values depending on where you approach it from. Let's define this more rigorously for on a 2D plane.

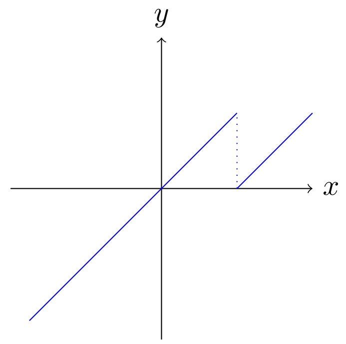
Definition: *(One-sided limit)* Let f be a function on $\mathbb{R}$ and let $a \in \mathbb{R}$ be a limit point of f. We say that the limit of f as $x \to a$ is one-sided if approaching it from the 'left side' of the limit yields a different result from a 'right-sided' approach. That is

$$\lim_{x\to a^+} f(x) \neq \lim_{x\to a^-} f(x) \tag{2.13}$$

where the $+$ and $-$ superscripts label the right- and left-sided approaches respectively.

Example: Let f be a piecewise function

$$f(x) = \begin{cases} x & \text{if } x < 1 \\ x - 1 & \text{if } x \geq 0 \end{cases} \tag{2.14}$$



Now consider the limit $\lim_{x\to 1^-} f(x)$. As you approach the limit point $a = 1$ from the 'left side' of f, you are approaching $f(x) = 1$.

$$\lim_{x\to 1^-} f(x) = 1 \tag{2.15}$$

However, if we approach from the other side, that is evaluate $\lim_{x\to 1^+} f(x)$, then you get 0.

$$\lim_{x\to 1^+} f(x) = 0 \tag{2.16}$$

Hence, $\lim_{x\to 1} f(x)$ is a one-sided limit.

2.2.4 Unbounded limits

The last two types of limits we must address are ones which are perfectly defined, but which diverge to $\pm\infty$. These often cause problems in many calculations and have famously plagued works in modern theoretical physics.

Definition: *(Unbounded limit)* Let $f(x)$ be a function on $\mathbb{R}$ with a limit point $a \in \mathbb{R}$. If the limit $\lim_{x\to a} f(x)$ deverges to ∞, we say that f is unbounded above a. Similarly, if this limit diverges to $-\infty$, then f is said to be unbounded below a.

Remark: We refrain from treating ∞ as a real number as this would lead to troubling contradictions. For example, $\infty + 1 = \infty$ would lead to $1 = 0$.

Example: Consider the function $f(x) = \exp(x)$ defined on $\mathbb{R}$. First, take the limit $\lim_{x\to 0} f(x)$. As you approach the limit point $a = 0$ the limit approaches $f(0) = \exp(0) = 1$.

$$\lim_{x\to 0} \exp(x) = 1 \tag{2.17}$$

Similarly, if consider $\lim_{x\to -\infty} \exp(x)$, then the limit evaluates to 0.

$$\lim_{x\to -\infty} \exp(x) = 0 \tag{2.18}$$

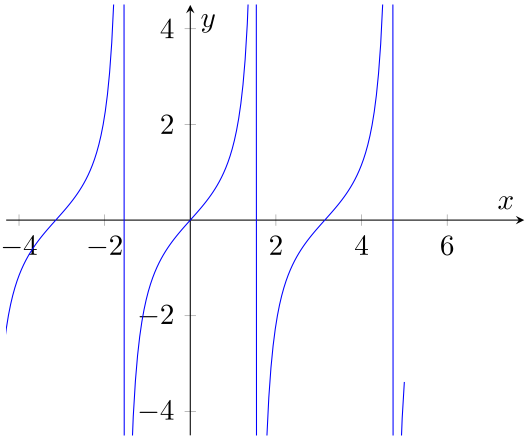
Therefore, both of these limits are bounded. However, let's now consider $\lim_{x\to\infty} \exp(x)$. This limit diverges to ∞.

$$\lim_{x\to\infty} \exp(x) = \infty \tag{2.19}$$

Hence, this particular limit is unbounded above the limit point.

There exist some functions which become unbounded both above and below the same limit point. However, for this to be true, the limit also has to be on-sided. This way, the limit will be unbounded below from one side of the limit point and above from the other side.

Example: Consider the function $f(x) = \tan(x)$ defined on $\mathbb{R}$.



First, we consider the limit as x approaches $\pi/2$ from $-\infty$. This limit evaluates to ∞ as $f(x) = \tan(x)$ is asymptotic to $x = \pi/2$ and grows without bound.

$$\lim_{x \to \frac{\pi}{2}^-} \tan(x) = \infty \tag{2.20}$$

Similarly, we now consider the right side of the limit point. That is, the limit as x approaches $\pi/2$ from positive ∞. As we said before, $f(x) = \tan(x)$ is asympototic to this limit point, however, it decays to $-\infty$ from the right side.

$$\lim_{x \to \frac{\pi}{2}^+} \tan(x) = -\infty \tag{2.21}$$

From this, it's important to note that due to the one-sidedness of this limit, we must explicitely state from which side we're approaching the limit point. As a consequence, we can say that the limit

$$\lim_{x \to \frac{\pi}{2}} \tan(x) = L \tag{2.22}$$

is undefined. Note that we didn't add a superscript to the limit point, and hence didn't specify from which side we're approaching the limit point, leaving L to be undefined.

2.2.5 Algebra of Limits

To round off this section and to give a bit more structure to the machinery of limits, we may define a kind of algebra on them. This will make it easier to work with and 'combine' limits in much the same way you would with other mathematical objects.

Definition: *(Algebra of limits)* Let f and g be functions with shared limit point a s.t. the limits $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exist. Then, if c is an arbitrary constant, the following four statements are true

- $\lim_{x\to a}(cf(x)) = c\lim_{x\to a} f(x)$ $\forall$ constants c
- $\lim_{x\to a}[f(x) + g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$
- $\lim_{x\to a}[f(x)g(x)] = (\lim_{x\to a} f(x))(\lim_{x\to a} g(x))$
- $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$ s.t. $\lim_{x\to a} g(x) \neq 0$

Remark: It's important to note the important subtlety of neglecting the possibility of one or both of these limits to evaluate to ∞.

Example: Consider the function $f(x) = 6\tan(x)/x$ defined on $\mathbb{R}\backslash\{0\}$. Notice that f is not defined at $x = 0$. However, we may compute

$$L = \lim_{x\to 0} \frac{6\tan(x)}{x} \tag{2.23}$$

By the algebra of limits, we can equate L to the product

$$L = 6\left(\lim_{x\to 0}\frac{1}{\cos x}\right)\left(\lim_{x\to 0}\frac{\sin(x)}{x}\right) \tag{2.24}$$

where we have used the fact that

$$\frac{\tan(x)}{x} = \frac{1}{\cos(x)} \cdot \frac{\sin(x)}{x} \tag{2.25}$$

Notice that both limits in the product evaluate to 1.

$$\lim_{x\to 0}\frac{1}{\cos x} = \lim_{x\to 0}\frac{\sin(x)}{x} = 1 \tag{2.26}$$

Hence, $L = 6 \cdot 1 \cdot 1 = 6$.

2.3 Operators

2.3.1 What is an Operator?

To make the next sections a little more straight forward, we will spend this section defining operators. There are, broadly speaking, two kinds of operators: linear and non-linear. First, however, let's define a general operator.

Definition: *(Operator)* An operator, $\hat{O}$, is a mapping of a space S, called the domain of $\hat{O}$, onto another space S^*, called the co-domain of $\hat{O}$, s.t. S and S^* may or may not be the same space.

Remark: It's a well established convention to mark operators with a hat $(\hat{\cdot})$ symbol. This is especially common in Quantum Mechanics where operators such as the Hamiltonian serve as the core of the theory.

The above definition is the closest possible general definition of operators. Mathematicians don't usually write such a broad definition of operators on account of how many different ways exist of interpreting such objects exist. For example, if we consider the space $\mathbb{R}$, then any function, f, of real variables is considered an operator with $S = S^* = \mathbb{R}$.

Example: Let $M \in \mathbb{R}^{n\times n}$ and let $S = \mathbb{R}^n$. Then, M is an operator on S with co-domain $S^* = S$.

We will mostly encounter operators which map spaces onto themselves. That is to say their domain and co-domain are the same space. However, the domain and co-domain are allowed to be different spaces. An example of this would be a matrix with complex entries acting on real vectors (example below).

Example: Let $M \in \mathbb{C}^{n\times n}$ and let $S = \mathbb{R}^n$. Then, M is an operator on S with co-domain $S^* = \mathbb{C}^n$.

The concept of operators is of major importance in calculus as we will be dealing with special operators (namely derivatives and integrals) which act over the space of functions. In fact, these two operators will turn out to be inverses of each other, which is, in essence, the statement of the fundamental theorem of Calculus. Therefore, we need to address the concept of inverse operators before we move on to the concept of linearity.

Definition: *(Inverse Operators)* Let $\hat{O}$ be an operator with domain S and let $f \in S$ be a member of S. Then, there exists an operator, $\hat{O}^{-1}$ called the inverse of $\hat{O}$ s.t.

$$\hat{O}\hat{O}^{-1}f = \hat{O}^{-1}\hat{O}f = f \tag{2.27}$$

for all $f \in S$. In other words, if $\hat{O}$ is the operator mapping S to the co-domain S^*, then $\hat{O}^{-1}$ is the operator mapping S^* onto S.

Example: Consider the operator

$$P = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix} \tag{2.28}$$

acting over $\mathbb{R}^2$. Then, the inverse of P, P^{-1} is the operator

$$P^{-1} = \frac{1}{6}\begin{pmatrix} 3 & 0 \\ -1 & 2 \end{pmatrix} \tag{2.29}$$

2.3.2 Linear Operators

There is a special class of operators called linear operators. Concepts associated with this class are especially important for our purposes in this chapter as we will later see (and prove) that both derivatives and integrals are examples of linear operators.

Definition: *(Linear operator)* Let $\hat{O}$ be an operator with domain S. Let also a, b be constants and $f, g \in S$ be members of the domain of $\hat{O}$. $\hat{O}$ is called a linear operator iff

$$\hat{O}(af + bg) = a\hat{O}f + b\hat{O}g \tag{2.30}$$

for all constants a, b and $f, g \in S$.

Example: Consider the space of real-valued $n \times n$ matrices $\mathbb{R}^{n\times n}$. Then, let $M \in \mathbb{R}^{n\times n}$ and let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$. In this case, $M : \mathbb{R}^n \to \mathbb{R}^n$, so M is an operator. In addition, for any such M acting on any linear combination of such $\mathbf{v}$ and $\mathbf{w}$, linearity holds.

$$M(a\mathbf{v} + b\mathbf{w}) = aM\mathbf{v} + bM\mathbf{w} \tag{2.31}$$

where a and b are constants. Therefore, if $M \in \mathbb{R}^{n\times n}$ then it is also a linear operator on $\mathbb{R}^n$.

2.4 Derivatives

2.4.1 Derivation from First Principles

We have now arrived at the first of the two topics which defines Calculus. This half is concerned with the study of rates of change. More specifically, derivatives are the tool used to find how fast a function is changing at an infinitely small point. To demonstrate this, imagine you get in a car and drive for 2 hours. If you cover 100km during this time, then you can say that your average speed was 50km/h. However, if I were to ask you how fast you were going exactly 45 minutes into your journey, you wouldn't be able to tell me. The best you could do is take some time around that point and calculate the average speed during that time. For example, you could say that you covered 1.2km between minute 44 and minute 46. Then, your average speed at minute 45 is roughly 36km/h. It's easy to see that the smaller you make this time period around minute 45, the more accurate your answer will be to how fast you were going at that instant in time.

This is exactly the way derivatives work. That being said, let's introduce this concept in a more formal way.

Theorem: *(Derivation from First Principles)* Let $f : \mathbb{R} \to \mathbb{R}$ be a smooth, continuous function. The immediate rate of change of the function (or gradient) at some point $a \in \mathbb{R}$ is called the derivative of f at the point a and is defined as

$$\frac{df}{dx}(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{2.32}$$

where $\frac{df}{dx}(a)$ marks the derivative of f at a.

Proof: Suppose we have a function f which is smooth and continuous over $\mathbb{R}$. The average gradient between two points $a, b \in \mathbb{R}$ s.t. $a < b$ is the ratio between the change in y over the change in x between these points.

$$m = \frac{f(b) - f(a)}{b - a} = \frac{\Delta f}{\Delta x} \tag{2.33}$$

If there is a non-zero difference, h, between a and b, then we can say that $b = a + h$ and so $b - a = h$. With this, we can rewrite the above equation in the form

$$\frac{\Delta f}{\Delta x} = \frac{f(a+h) - f(a)}{h} \tag{2.34}$$

Consider decreasing the difference between a and b to the point where the value h is infinitesimally small. In this case, the point b will approach the point a until they are essentially the same point. This, consequently, will give us the average gradient of f in an interval which is infinitesimally small around the point a. In turn, this tells us the rate of change of f at exactly the point $x = a$. Mathematically, this may be expressed as a limit of $\Delta f / \Delta x$ as $h \to 0$.

$$\frac{\Delta f}{\Delta x}(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{2.35}$$

This is called the derivative of f and this is marked by replacing Δ with d to signify an infinitesimal change. Hence, we arrive at the equation

$$\frac{df}{dx}(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{2.36}$$

Q.E.D.

Example: Consider the function $f(x) = x^2$. We can compute the derivative of this function by employing the above theorem.

$$\frac{df}{dx} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{(x+h)^2 - x^2}{h} \tag{2.37}$$

This reduces to

$$\frac{df}{dx} = \lim_{h\to 0} \frac{x^2 + h^2 + 2xh - x^2}{h} = \lim_{h\to 0}(h + 2x) \tag{2.38}$$

which is a straight forward limit. Hence

$$\frac{df}{dx} = 2x \tag{2.39}$$

Example: Consider the function $f(x) = \sin(x)$. We can compute the derivative of this function by employing the above theorem.

$$\frac{df}{dx} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{\sin(x+h) - \sin(x)}{h} \tag{2.40}$$

Recall the identity $\sin(A+B) = \sin(A)\cos(B) + \sin(B)\cos(A)$. We can use this to rewrite the limit as

$$\frac{df}{dx} = \lim_{h\to 0} \frac{\sin(x)\cos(h) + \sin(h)\cos(x) - \sin(x)}{h} \tag{2.41}$$

which is can then be written as

$$\frac{df}{dx} = \lim_{h\to 0}\left(\frac{\cos(h) - 1}{h}\sin(x) + \frac{\sin(h)}{h}\cos(x)\right) \tag{2.42}$$

Notice that $\sin(h)/h$ approaches 1 as $h \to 0$. Notice also that since $\cos(0) = 1$, $\cos(h)/h$ approaches $1/h$ as $h \to 0$. Hence, $(\cos(h) - 1)/h$ approaches 0 as $h \to 0$. Therefore, the limit evaluates to

$$\lim_{h\to 0}\left(\frac{\cos(h) - 1}{h}\sin(x) + \frac{\sin(h)}{h}\cos(x)\right) = 0 \cdot \sin(x) + 1 \cdot \cos(x) \tag{2.43}$$

hence

$$\frac{df}{dx} = \cos(x) \tag{2.44}$$

Remark: This technique is called *Derivation from First Principles* as we are using the formula first introduced by Cauchy and Weierstrass to compute derivatives. It would be very tedious to compute all derivatives this way, and so, we luckily have a wide variety of useful rules and shortcuts we can employ. We will see these later on.

2.4.2 The Derivative as a Linear Operator

We have talked about the concept of operators and, more specifically, linear operators in the previous section. At the end of the section, we touched on the fact that derivatives and integrals are examples of such objects. We will now use derivation from first principles to prove this for derivatives.

Remark: The notation we have previously used to write derivatives needs a slight 'unpicking' to clear some things up. We have so far written the derivative of a function f with respect to x as

$$\frac{df}{dx} \tag{2.45}$$

This, however, needs to be seen as an operator acting on a function to produce another function. So, we can write the derivative in this way

$$\frac{d}{dx}f = f' \tag{2.46}$$

where the function f is acted on by the derivative operator d/dx to produce the derivative of f, f'. A common shorthand for this is $d_x f$ or $\partial_x f$ for partial derivatives which we will cover later.

To prove that derivatives are linear operators, we must first check if they're operators at all. For this to be the case, we recall our definition of an operator from Section 3.3.1: *an operator is an object which maps a space S (domain) to another space S^* (co-domain) s.t. S and S^* may or may not be the same space.* Hence, we have to ask what is the domain of derivatives with respect to a real variable x? Take a second to think about this...

The correct answer is the space of functions over $\mathbb{R}$ i.e. the space of functions f s.t. $f : \mathbb{R} \to X$ where X is any vector-space. We denote this space of arbitrary functions $F(\mathbb{R}, X)$. So, for a derivative to be considered an operator, we need it to map $F(\mathbb{R}, X)$ to some other space S^* consistently.

$$\frac{d}{dx} : F(\mathbb{R}, X) \to S^* \tag{2.47}$$

Remark: Note that $a \in F(\mathbb{R}, X)$ for all constants a.

But what is this space S^*? Let's turn to our definition of derivatives.

$$\frac{d}{dx}f = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \tag{2.48}$$

Since $f(x) \in F(\mathbb{R}, X)$, it also follows that $f(x+h) \in F(\mathbb{R}, X)$. To see this, we must realise that changing the argument of f from x to $x+h$ doesn't change the fact that f takes real arguments. This is because h is also a real variable. Hence, $x+h$ is also real. The only change we bring with this transformation is a shifting of the function along $\mathbb{R}$. In addition, since $f(x)$ and $f(x+h)$ are both members of $F(\mathbb{R}, X)$, then so is $f(x+h) - f(x)$. If we let $f(x+h) - f(x) = g(x+h)$ then we have

$$\frac{d}{dx}f = \lim_{h\to 0} \frac{g(x+h)}{h} \tag{2.49}$$

s.t. $g(x+h) \in F(\mathbb{R}, X)$. The last important thing to notice is that since g is a member of $F(\mathbb{R}, X)$ and we are taking the limit of a real variable $h \to 0$, then as long as the limit is well defined,

$$\lim_{h\to 0} \frac{g(x+h)}{h} \in F(\mathbb{R}, X) \tag{2.50}$$

and so $\frac{d}{dx}f \in F(\mathbb{R}, X)$. In other words, the derivative with respect to a real variable x maps $F(\mathbb{R}, X)$ to itself. Hence $S^* = F(\mathbb{R}, X)$ and we can write

$$\frac{d}{dx} : F(\mathbb{R}, X) \to F(\mathbb{R}, X) \tag{2.51}$$

which declares the derivative an operator.

The last thing to do then, is to check whether or not the derivative is linear. Again, let's look at our equation for derivation. For derivation to be linear we need the relation

$$\frac{d}{dx}(af + bg) = a\frac{df}{dx} + b\frac{dg}{dx} \tag{2.52}$$

to be true for all constants a, b and all functions $f, g \in F(\mathbb{R}, X)$. Hence let's rewrite the derivative in the following way

$$\frac{d}{dx}(af + bg) = \lim_{h\to 0} \frac{af(x+h) + bg(x+h) - af(x) - bg(x)}{h} \tag{2.53}$$

Notice that this can be rearranged into the form

$$\frac{d}{dx}(af + bg) = \lim_{h\to 0} \frac{a[f(x+h) - f(x)] + b[g(x+h) - g(x)]}{h} \tag{2.54}$$

Now we can use the algebra of limits to simplify this and write

$$\frac{d}{dx}(af + bg) = a \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} + b \lim_{h\to 0} \frac{g(x+h) - g(x)}{h} \tag{2.55}$$

The two limits in this form then correspond to the derivatives of f and g respectively by first principles. Hence

$$\frac{d}{dx}(af + bg) = a\frac{df}{dx} + b\frac{dg}{dx} \tag{2.56}$$

and so linearity of the derivative is proven. Therefore, we have proven that derivatives with respect to a real variable are linear operators that map $F(\mathbb{R}, X)$ onto itself.

2.4.3 Common Derivatives

As mentioned before, derivatives are generally not computed through first principles. Instead, there are a number of common rules which help speed up differentiation of functions. In this section, we'll look at and prove some of these rules. First, let's look at derivatives of polynomials.

Rule I: (Derivative of Polynomials) Suppose you have a polynomial of degree n.

$$p_n(x) = \sum_{i=0}^{n} a_i x^i \tag{2.57}$$

Then, the derivative of p_n is

$$\frac{dp_n}{dx} = \sum_{i=0}^{n} a_i i x^{i-1} \tag{2.58}$$

Rule II: (Derivatives of Trigonometric Functions) The derivatives of the two basic trigonometric functions are

$$\frac{d}{dx}\sin(x) = \cos(x) \tag{2.59}$$

$$\frac{d}{dx}\cos(x) = -\sin(x) \tag{2.60}$$

Remark: Derivatives of other trig functions can be found using rules from the next few sections.

Rule III: (Derivatives of Logarithms) The derivative of a logarithm of base b is

$$\frac{d}{dx}\log_b(x) = \frac{1}{x\ln(b)} \tag{2.61}$$

Hence, there is a special case for the natural logarithm: $d\ln x/dx = 1/x$.

2.4.4 Chain Rule

The first of three rules concerning the mechanism of differentiation deals with the derivatives of function compositions. Suppose we have a composition of two functions f and g, $f(g(x))$. How would we find the derivative of f with respect to x if there is another function 'between' f and the variable x? In other words, how do we compute df/dx? If we treat derivatives as fractions, it's quite easy to think of a strategy. We can find dg/dx easily since that's a direct derivative. Similarly, if we treat g as a variable, we can find the derivative df/dg directly. Now, we can find the product of these two derivatives.

$$P = \frac{df}{dg} \cdot \frac{dg}{dx} \tag{2.62}$$

Notice that this product simplifies to give df/dx. This is precisely the derivative we are after and the strategy can be formalised in the following theorem.

Theorem: *(Chain Rule)* Let f and g be functions over $\mathbb{R}$ such that the co-domain of g, S_g satisfies $S_g \subseteq \mathbb{R}$. Then, the function composition $h(x) = (f \circ g)(x) = f(g(x))$ exists and its derivative is

$$\frac{d}{dx} f(g(x)) = \frac{df}{dg} \cdot \frac{dg}{dx} \tag{2.63}$$

Example: Consider the functions $f(x) = \sin(x)$ and $g(x) = x^2$ over $\mathbb{R}$. Then the composition $f \circ g$ is $h(x) = \sin(x^2)$ exists and its derivative is

$$\frac{dh}{dx} = \frac{d}{dx}(x^2) \cdot \frac{d}{dg}(\sin(g)) \tag{2.64}$$

which is equivalent to

$$\frac{dh}{dx} = 2x \cdot \cos(g) = 2x\cos(x^2) \tag{2.65}$$

The composition $g \circ f$ also exists: $g(f(x)) = H(x) = \sin^2(x)$. We can compute its derivative similarly

$$\frac{dH}{dx} = \frac{d}{df}(f^2) \cdot \frac{d}{dx}(\sin(x)) \tag{2.66}$$

which evaluates to

$$\frac{dH}{dx} = 2f \cdot \cos(x) = 2\sin(x)\cos(x) \tag{2.67}$$

Example: Consider the functions $\ln : \mathbb{R}^+ \to \mathbb{R}$ and $\cos : (0, \pi/2) \to (0, 1)$. Then the composition $f \circ g$ is $h(x) = \ln(\cos(x))$ exists over $(0, \pi/2)$ and its derivative is

$$\frac{dh}{dx} = \frac{d}{dg}(\ln(g)) \cdot \frac{d}{dx}(\cos(x)) \tag{2.68}$$

which is equivalent to

$$\frac{dh}{dx} = \frac{1}{g} \cdot (-\sin(x)) = -\frac{\sin(x)}{\cos(x)} \tag{2.69}$$

Notice that this is the same as $-\tan(x)$. Hence

$$\frac{d}{dx} \ln(\cos(x) = -\tan(x) \tag{2.70}$$

Remark: There is a deeper reason why this theorem holds, however, for our purposes, this explanation will suffice.

2.4.5 Product, Leibniz and Quotient Rules

The following rules, as the name might suggest, help us compute derivatives of products and ratios of functions. Given two functions f and g, we may compute the derivative of $f(x)g(x)$ if we know df/dx and dg/dx. The first theorem deals with only computing the first-order derivative of such a product.

Theorem: *(Product Rule)* Let f and g be functions over $\mathbb{R}$. Also let $h(x) = f(x)g(x)$. Then the first-order derivative of h is

$$\frac{dh}{dx} = \frac{df}{dx}g(x) + f(x)\frac{dg}{dx} \tag{2.71}$$

Example: Consider the functions $f(x) = x$ and $g(x) = e^x$ and let $h(x) = xe^x$. Then the derivative of h is

$$\frac{dh}{dx} = \frac{d}{dx}(x) \cdot e^x + x \cdot \frac{d}{dx}e^x \tag{2.72}$$

since e^x is its own derivative, we have

$$\frac{dh}{dx} = e^x + xe^x = e^x(1+x) \tag{2.73}$$

If we wish to compute the n^{th} order derivative of a product, we may use an extension of the above theorem called the *Leibniz Rule*. This theorem extends the product rule to functions in the space of n-times continuously differentiable functions over $\mathbb{R}$, $C^n(\mathbb{R})$. This means that if $f \in C^n(\mathbb{R})$ then all derivatives of f exist up to and including order n and are all continuous.

Theorem: *(Leibniz Rule)* Let $f, g \in C^n(\mathbb{R})$. Also let $h(x) = f(x)g(x)$. Then $h \in C^n(\mathbb{R})$ and the n^{th} derivative of h is the sum

$$\frac{d^n h}{dx^n} = \sum_{i=0}^{n} \frac{n!}{i!(n-i)!}\frac{d^i f}{dx^i}\frac{d^{n-i} g}{dx^{n-i}} \tag{2.74}$$

Or, with more compact notation

$$h^{(n)}(x) = \sum_{i=0}^{n} \binom{n}{i} f^{(i)}(x) g^{(n-i)}(x) \tag{2.75}$$

where $\binom{n}{i} = \frac{n!}{i!(n-i)!}$.

Remark: From this point on, we will mostly stick to the above notation for derivatives. i.e.: $f^{(i)}(x)$ will mean the i^{th}-order derivative of the function f with respect to x. We will sometimes break away from this, for example when dealing with partial derivatives. It is also important to note that while $f^{(i)}(x)$ is an i^{th}-order derivative, $f^i(x)$ is an i^{th}-order exponent.

Example: Consider the functions $f(x) = x^3$ and $g(x) = e^{2x}$ and let $h(x) = x^3e^{2x}$. Let us compute the 3^{rd}-order derivative of h. First we compute the derivatives up to and including order 3 of both f and g separately.

$$f^{(1)}(x) = 3x^2,\ f^{(2)}(x) = 6x,\ f^{(3)}(x) = 6 \tag{2.76}$$

$$g^{(1)}(x) = 2e^{2x},\ g^{(2)}(x) = 4e^{2x}, g^{(1)}(x) = 8e^{2x} \tag{2.77}$$

Applying the Leibniz rule, we can compute $h^{(3)}(x)$

$$\begin{aligned} h^{(3)}(x) &= f^{(3)}g + 3f^{(2)}g^{(1)} + 3f^{(1)}g^{(2)} + fg^{(3)} \\ &= 6e^{(2x)} + 3(6x)(2e^{2x}) + 3(3x^2)(4e^{2x}) + x^3(8e^{2x}) \\ &= 2e^{2x}(3 + 18x + 18x^2 + 4x^3) \end{aligned} \tag{2.78}$$

Conversely to the product rule, there is a rule that allows us to evaluate derivatives of ratios of functions. This, very creatively, is called the *Quotient Rule* and is formally written as the following theorem.

Theorem: *(Quotient Rule)* Let f and g be functions over $\mathbb{R}$ s.t. $f^{(1)}$ and $g^{(1)}$ exist and g is non-zero. If $h = f/g$ then the derivative of h is

$$h^{(1)}(x) = \frac{g(x)f^{(1)}(x) - f(x)g^{(1)}(x)}{g^2(x)} \tag{2.79}$$

Remark: Note that the ratio of the first derivatives of f and g are generally not equal to the first derivative of the ratio of f and g.

$$\frac{f^{(1)}(x)}{g^{(1)}(x)} \neq \left(\frac{f(x)}{g(x)}\right)^{(1)} \tag{2.80}$$

This is a common mistake made by first- and even some second-year physics students which can lead to nonsensical results such as a rod cooling off the more heat you pass onto it. (This was an answer obtained by a friend of mine in a thermodynamics seminar due to this exact oversight.) For this to be the case, the following non-linear differential equation must hold

$$\frac{g(x)}{g^{(1)}(x)}\left(1 - \frac{g(x)}{g^{(1)}(x)}\right) f^{(1)}(x) = f(x) \tag{2.81}$$

Example: Let f be the function.

$$f(x) = \frac{\sin(x)}{x} \tag{2.82}$$

Since f is a quotient of two functions $\sin(x)$ and x, we can use the quotient rule to compute its first derivative.

$$f^{(1)}(x) = \frac{x\cos(x) - \sin(x)}{x^2} = \frac{\cos(x)}{x} - \frac{\sin(x)}{x^2} \tag{2.83}$$

Example: Let f be the function.

$$f(x) = \frac{\ln(x)}{\cos(x)} \tag{2.84}$$

Since f is a quotient of two functions $\ln(x)$ and $\cos(x)$, we can use the quotient rule to compute its first derivative.

$$f^{(1)}(x) = \frac{\cos(x)/x + \ln(x)\sin(x)}{\cos^2(x)} = \frac{1}{x\cos(x)} + \frac{\ln(x)}{\cos(x)}\tan(x) \tag{2.85}$$

2.4.6 Derivatives of Inverse Functions

We have now seen quite a few different methods of evaluating derivatives of functions. However, it is also worth knowing how the inverse of a function changes. This section may be viewed as an aside as this is not a major part of what you might see in most Calculus modules. It is, however, needed to be a well-rounded and complete student of the subject and may sometimes simplify calculations in research.

If we have a function f s.t. $f(y) = x$ then its inverse will be the function $f^{-1}(x) = y$. Assuming that $f^{(1)}$ exists, we may write

$$\frac{df}{dy} = \frac{dx}{dy} \tag{2.86}$$

Notice that $dx/dy = (dy/dx)^{-1}$. Hence, we can write

$$\frac{df}{dy} = \left(\frac{dy}{dx}\right)^{-1} \implies \frac{dy}{dx}\frac{df}{dy} = 1 \tag{2.87}$$

and since $y = f^{-1}(x)$, we have an equation for the derivative of the inverse.

$$\frac{d}{dx}f^{-1}(x) = \frac{1}{f^{(1)}(y)} \tag{2.88}$$

2.4.7 Parametric Differentiation

2.4.8 Implicit Differentiation

2.4.9 L'Hopital's Rule

2.5 Integrals

Chapter 3

Ordinary Differential Equations

Chapter 4

Vector Calculus

Introduction

This chapter covers another foundational topic in mathematics, physics and engineering. In simple terms, vector calculus is literally concerned with computing the derivatives and integrals of vectors. The reason why this topic is critical to understand is because in the real world most of everything is governed by different forces, which are vectors as they have direction and magnitude, so having a technique that allows you to analyse the different effects these forces have when interacting with each other is highly useful.

4.1 Basic properties of vector calculus

4.1.1 What is ∇

∇ put very simply is an operator. Mathematically it is presented as such:

$$\nabla = \hat{\mathbf{i}}\frac{\partial f}{\partial x} + \hat{\mathbf{j}}\frac{\partial f}{\partial y} + \hat{\mathbf{k}}\frac{\partial f}{\partial z} \tag{4.1}$$

As you can see, the operator ∇ is defined as a vector whose components are partial derivative operators along the standard basis vectors $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$

Chapter 5

Classical Mechanics

5.1 Newton's Laws

5.2 Kinematics

5.3 Conservative forces

Chapter 6

Hamiltonian and Lagrangian Dynamics